

An Analytical Framework for Error and Hallucination in Deployed Generative AI Systems

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Abstract

Evaluation of errors in a deployed generative AI system presents two related problems. First, the information used to produce a response may differ from the information that should govern its evaluation. Second, identifying an error does not establish how the configured system produced it. This paper develops an authority-first framework that addresses these problems in sequence. For each assessable assertion, the framework identifies the governing reference and classifies the assertion as supported, contradicted, unsupported, or unresolved. In parallel, it compares the delivered response with applicable task obligations. These two evaluative branches support two related error classifications: hallucination, which concerns a representational claim contradicted by or lacking support required from its governing reference, and task-obligation fidelity error, which concerns failure to satisfy an applicable requirement. The framework then separates error classification from localization and causal attribution, and error occurrence from consequence. Applications to temporal change, continuing-project state, retrieval-augmented generation, and propagation show why evaluation and control selection should concern a configured system performing a specified task rather than a model name alone. The framework is conceptual, and its practical and empirical value remains to be tested.

Key words: AI governance; AI evaluation; model error; hallucination; factuality; faithfulness; governing reference; task-obligation fidelity; project-state fidelity; RAG

A. Introduction

Generative AI applications increasingly operate as *configured systems* rather than as stand-alone models. We use *configured system* to mean the trained model together with the task-specific deployment resources, processes, settings, tools, and controls that help produce the response delivered to the user. Such a system may include an organization’s documents, retrieved information, software tools, prior interactions, and technical or procedural controls. When work continues across time, the response may also depend on decisions and records created during earlier work.

This broader system setting creates two evaluation problems. The first arises when the information used to produce a response is not the information that should govern its evaluation. A system may, for example, receive an outdated document when a current rule governs, or rely on general knowledge when the task requires fidelity to a designated source. Evaluation of the response would then be misleading unless the applicable reference is identified before the response is judged.

The second problem arises because an observed error does not, by itself, explain how the error was produced. A false claim may reflect a limitation of the trained model, omission of relevant information

during context construction, failure during generation to use supplied information, or transformation of the response before delivery. Moreover, the same production problem may lead to different outcomes. Failed retrieval may be followed by abstention in one case and fabrication in another. Identification of an error, localization of its possible source, and attribution of its cause should therefore be treated as separate questions.

Calling a response error a hallucination does not resolve either problem. The hallucination and faithfulness definitions cited here emphasize, in varying degrees, factual falsehood, lack of evidentiary support, inconsistency with supplied material, the distinction between contradiction and unverifiable addition, and the distinction between source faithfulness and external factuality ([1]–[7]). These cited definitions and distinctions are useful once the reference applicable to a claim has been established. They do not, by themselves, determine which reference should govern the claim.

This paper begins with that question. We use the term *authority-first evaluation* for a sequence that begins by selecting the governing reference and then proceeds to discrepancy classification, localization and causal attribution, consequence assessment, and governance response. The sequence applies after a response has been delivered. It does not assume that every deficiency originated in the trained model, and it does not require a causal conclusion when the available evidence establishes only that a discrepancy occurred.

The framework also separates hallucination from task-obligation fidelity. Task-obligation fidelity means conformity with requirements established by a governing source, instruction, contextual record, or authorized project state. By authorized project state, we mean the accepted decisions, requirements, and statuses governing continuing work at the time of the task. An approved requirements document may therefore form part of the authorized state, whereas an unapproved proposal or a superseded draft does not. A task-obligation fidelity error may be entirely behavioral, such as exceeding a word limit. It may also overlap with hallucination if the response falsely represents an instruction, source, record, or project decision.

The analysis is conceptual rather than empirical. We organize it around four questions: What governs a claim? What discrepancy occurred? What can the available evidence establish about how the discrepancy was produced? What response is appropriate? Section B describes how a deployed system produces a delivered response. Section C develops governing-reference selection and discrepancy evaluation. Section D distinguishes hallucination from task-obligation fidelity, and Section E separates error-source localization from causal attribution. Section F applies the framework to time, continuing-project state, and retrieval-augmented generation. Section G considers consequences, controls, limitations, and conclusions.

B. Response Production in a Deployed System

We begin with the production sequence that underlies the later evaluation. At interaction time t , the configured system receives the current production input x_t , including the user’s request and any applicable instructions supplied with it. The system also has access to prior interaction history H_{t-1} and may draw on external resources E_t , including documents, databases, tools, and current observations. A context-

construction process R retrieves, selects, compresses, orders, or labels information to produce the operative context:

$$C_t = R(x_t, \mathcal{H}_{t-1}, E_t). \quad (1)$$

We use *operative context* to mean the supplementary information C_t actually presented for generation. It is not the same as all information accessible to the system. Relevant material may remain outside the operative context because it was not retrieved, was displaced by other material, or lost an important qualification during compression. Nor does inclusion ensure effective use: model performance may depend on where relevant information appears in a long context ([8]). The current input x_t guides context construction and is also supplied directly to generation. It is a production input, whereas the task specification q_t introduced in Section C is an evaluative representation of what the delivered response was required to do.

After the operative context has been constructed, let $M(t_0)$ denote the trained model used at response time t . The marker t_0 represents the model’s training-information boundary in simplified form; it is not a claim that all training information shares one exact date. Conditional on the current input and operative context, the deployed generation process produces a candidate response:

$$\tilde{y}_t \sim P_{M(t_0)}(\cdot | x_t, C_t). \quad (2)$$

Equation (2) summarizes generation; it does not specify a decoding algorithm. The model, current input, and operative context determine conditional predictions, while decoding and successive selections realize one response path. Information available after t_0 may be absent from the trained model, but that absence does not by itself determine the delivered response. Current information, tools, decoding, or later controls may correct, transmit, or amplify an initial limitation.

Generation need not be the final stage. Before delivery, a post-generation process may verify, qualify, redact, rewrite, refuse, or block the candidate response. Let Σ denote the applicable rules and controls, and let I_Σ denote the process that implements them:

$$y_t = I_\Sigma(\tilde{y}_t, x_t, C_t). \quad (3)$$

The delivered response y_t may therefore differ from the candidate response. This distinction becomes material if a control removes a necessary qualification, inserts an explanation, or prevents an erroneous claim from reaching the user.

Together, Equations (1)–(3) allow us to distinguish four objects relevant to evaluation: information potentially accessible to the system, the operative context actually supplied, the candidate response generated, and the response ultimately delivered. Their purpose is to show why a deficiency in the delivered response should not automatically be attributed to the trained model; they are not a causal model of response error.

C. Governing References and Response Evaluation

Given a delivered response to a task, evaluation first requires identification of the claims that can be assessed. Following the claim-level approach used in factuality evaluation ([9]), we call each such claim

an *assertion*: an explicit or reasonably implied claim whose relation to an applicable reference can be evaluated. A response may contain several assertions, and different assertions in the same response may require different governing references.

Evaluation of an assertion requires an explicit account of the task. Let q_t denote the evaluative task specification reconstructed from x_t together with any controlling instructions and authorized contextual or project state applicable to the delivered response. Thus x_t records what entered production, while q_t states what the response was required to do for evaluation. Let Π_t denote the authority policy that assigns priority among candidate references when more than one appears relevant. For a source-faithful summary, for example, Π_t may specify that the designated source governs even if external knowledge conflicts with it. Establishing the task specification and authority policy before examining a preferred answer reduces the risk of selecting a convenient standard after the fact.

A simple conflict illustrates how these inputs work. Suppose a task asks the system to apply the organization’s currently authorized policy, while the available materials include a superseded manual, a later approved manual, and a user message quoting the older rule. The task specification identifies the current-policy task; provenance, temporal applicability, and authorization status distinguish the materials; and an explicit priority rule in Π_t —such as giving the currently approved manual priority over superseded material—selects the governing reference. If the task instead asks what the earlier manual required, q_t changes the temporal target and may change the result. When no applicable authority policy exists, or when the relevant priority rule is materially contested, evaluation is not yet well-posed: the evaluator should document the conflict and resolve or stipulate the policy before classifying the assertion-reference relation.

More formally, let a_{tj} denote assertion j in the delivered response, let k_{tj} denote its evaluative role, and let $\mathcal{A}_{tj}$ denote the set of candidate references potentially applicable to the assertion. If ρ is the reference-selection rule, the governing reference is:

$$r_{tj}^* = \rho(a_{tj}, q_t, k_{tj}, \mathcal{A}_{tj}; \Pi_t). \quad (4)$$

The governing reference is an evaluative standard; it need not be information that the model received. An assertion may agree with the operative context and still conflict with the governing reference. Conversely, it may be correct as a statement about the external world and still fail to represent a designated source faithfully. The roles in Table 1 are evaluative roles rather than mutually exclusive claim types. When one assertion performs more than one role, each relevant relation should be evaluated separately unless Π_t establishes which role governs the task.

Table 1. Assertion roles and governing references

Assertion role	Potential governing reference	Illustration
External factual claim	Current verifiable information at the time relevant to the task	Who currently holds an office; a current price or rule

Assertion role	Potential governing reference	Illustration
Designated-source claim	Designated source, including its provenance and qualifications	What a supplied paper states
Claim about an applicable instruction	Applicable instruction	Whether the task required a specified scope, format, or prohibited transformation
Contextual claim	Authoritative contextual record	What was established earlier in the operative exchange
Project-state claim	Authorized project state	Whether a proposal was accepted, revised, or superseded

Once the governing reference has been selected, the assertion can be compared with it. Let v denote the comparison rule, with q_t and Π_t specifying the task and authority parameters that determine the support required. Let V_{tj} denote the resulting classification:

$$V_{tj} = v(a_{tj}, r_{tj}^*; q_t, \Pi_t), \quad (5)$$

$$V_{tj} \in \{\text{supported}, \text{contradicted}, \text{unsupported}, \text{unresolved}\}.$$

We call an assertion supported when the governing reference supplies the support required by the task, and contradicted when the reference conflicts with the assertion. Unsupported applies only when the governing reference has been adequately identified and is sufficiently available for evaluation, yet it does not supply support required by the task. For example, a complete designated report may contain no basis for an added numerical claim. Unresolved applies when unavailable or incomplete reference material, ambiguity, or other evidentiary limits prevent the evaluator from deciding whether the assertion is supported, contradicted, or unsupported. A missing appendix needed to evaluate the same numerical claim would produce an unresolved relation rather than an unsupported one. An unresolved classification does not establish that the assertion is supported and should not be treated as a passing result.

For response features that are not assertions, we use a parallel branch for task obligations. The evaluator identifies each applicable obligation from q_t and Π_t , identifies its governing requirement, and compares the delivered response with that requirement. The result records whether the obligation was satisfied, violated, or cannot be resolved from the available evidence. We express this conformance assessment in prose because it does not require a second claim-reference equation.

A formatting violation, such as exceeding a word limit, can therefore be evaluated directly against its requirement without being treated as a representational claim. It falls outside Equation (5) but enters the task-obligation branch. Together, the assertion-reference and obligation-conformance branches provide the evaluative basis for the two error classifications developed in Section D.

D. Hallucination and Task-Obligation Fidelity

The framework applies two error classifications to a delivered response. Hallucination concerns the representational content of an assertion. Task-obligation fidelity error concerns failure of the response to

conform to an applicable requirement. Because a response may both misrepresent a task requirement and fail to satisfy that requirement, the two classifications may apply separately or together.

More specifically, a representational assertion may concern an external fact, a designated source, an applicable instruction, an authoritative contextual record, or an authorized project state. We classify the assertion as a hallucination if it is contradicted by, or lacks support required from, its governing reference. Hallucination is therefore a property of a delivered claim assessed after the fact relative to its governing reference. The classification does not, however, establish how the claim was produced.

This is the paper’s proposed *umbrella usage* of hallucination. Narrower usages often focus on external factuality or fidelity to a designated source, and evaluators may retain labels such as factual error, source-unfaithful claim, instruction misrepresentation, or project-state error for particular domains. Those narrower labels remain compatible with the authority-first sequence: the evaluator still identifies the governing reference and classifies the claim-reference relation. The umbrella usage does not convert a purely behavioral nonconformance, such as exceeding a word limit without making a claim about compliance, into hallucination.

A task-obligation fidelity error, by comparison, occurs when the delivered response departs from a requirement established by a governing source, instruction, contextual record, or authorized project state. The requirement may govern the form of the response, such as its word limit, or require faithful representation of a source.

Consider, for example, a request for a source-faithful summary of no more than 200 words. An accurate summary of 300 words violates the length requirement and therefore contains a task-obligation fidelity error; the excess length does not, by itself, constitute hallucination. If the response also states that it complies with the 200-word limit, that statement is a representational assertion contradicted by the instruction. In the latter scenario, the response would therefore exhibit both hallucination and task-obligation fidelity error.

Allowing the two error classifications to overlap avoids two distortions. If every task failure were called hallucination, a formatting violation and a fabricated citation would be placed in the same category. If false claims about instructions, contextual records, or project state were excluded from hallucination merely because they concern different kinds of governing references, representational claims with the same relation to their governing references would be classified differently. The relevant question for hallucination classification is whether the response makes a representational claim and, if so, how that claim relates to its governing reference.

These error classifications establish what occurred, not why it occurred. A hallucination may be identified even when no production record reveals how the claim arose. Conversely, a retrieval or tool failure is not itself a hallucination if the system abstains and makes no discrepant assertion. This separation allows possible production causes to be investigated without placing an assumed cause inside the definition of the error.

E. Localization and Causal Attribution

Once a response error has been identified and classified, evaluation may turn to its production. Two questions then arise: Where could the error have arisen, and what evidence shows that a condition at that location contributed to it? We use localization for identifying a plausible production locus or set of loci, and causal attribution for establishing that a condition or process at a locus contributed to the delivered error. Section B identifies four principal places to investigate: the trained model, context construction, generation and decoding, and post-generation processing. To relate these concrete loci to broader production processes, we group them into three non-exhaustive mechanism families: Model Approximation, Autoregressive Realization, and Deployment Mediation. This grouping provides a common structure for considering how conditions at different loci may contribute separately or interact.

We map Model Approximation to model formation and the trained model $M(t_0)$, where errors may be inherited from forming a finite predictive model from bounded information through design-contingent choices. Autoregressive Realization maps to Equation (2), where conditional prediction, decoding, and earlier selections shape the candidate response. Deployment Mediation maps to the external resources and context-construction process in Equation (1), as well as the tools, interaction history, and post-generation process represented in Equation (3). These mappings identify where a production investigation may look. Because the families may overlap, none constitutes a causal finding inferred from the delivered error. A separate paper by this author develops this grouping in greater detail ([10]), but that paper is not required to apply the distinctions used here. Research in knowledge-grounded conversational settings also shows that hallucination can reflect both dataset and model contributions ([11]).

Consider a hallucination about a designated source. The relevant passage may not have been retrieved, may have been omitted during context construction, or may have entered the operative context but not been used during generation. The candidate response may instead have represented the source accurately before a post-generation process altered the claim. Each path can produce the same delivered hallucination, and more than one path may contribute. The response and governing source establish the error; they do not identify its production history.

Distinguishing among plausible production paths requires production evidence. Retrieval and context-construction logs may show whether relevant material entered C_t . A stored candidate response may show whether post-generation processing changed the claim. Repeated trials may reveal sensitivity to realization, while controlled interventions may test whether changing a production condition changes the error. Such records establish observed conditions; causal attribution additionally requires evidence that a condition contributed to the delivered error. When records are incomplete, the evaluator may identify several plausible loci without attributing the error to any one of them.

Table 2 applies this evidentiary distinction to several response-production conditions. Its production loci are nonexclusive: the listed locus is a plausible place to investigate, not a cause inferred from the response-level classification. The evidence column indicates what would be needed to move from an observed condition toward a supported production finding.

Table 2. Error occurrence, possible production locus, and evidence for causal attribution

Observed condition	Response-level classification	Nonexclusive production inquiry and evidence
Retrieval returns no relevant material and the system abstains	No discrepant assertion; whether an obligation error occurred depends on the task	Retrieval logs establish the missing result. Because no response error is established on these facts alone, there is no error cause to attribute.
Relevant source material is not retrieved and the response invents a source-backed answer	Hallucination involving a designated source	Deployment Mediation and Autoregressive Realization may interact; retrieval, context, and generation records are needed to test contribution.
Accurate source material is present in the operative context, but the response contradicts it	Hallucination involving a designated source	Autoregressive Realization, Model Approximation, or later mediation may contribute; repeated trials, model comparisons, and candidate/delivered records can help distinguish them.
A current factual answer is contradicted by the current governing reference	Factual hallucination	Model Approximation, absent or obsolete context, realization, and propagation are plausible; training boundary, context records, and controlled comparisons are needed.
A post-generation process inserts an unsupported explanation	Hallucination if the explanation lacks required support from its governing reference	Deployment Mediation; comparison of candidate and delivered responses directly establishes the transformation, while surrounding records test why it occurred.

We can therefore draw a bounded conclusion from this analysis. The delivered response and governing standard may establish what error occurred. Production records and interventions determine how far the evaluator can go in locating or attributing it. Consequence and control choice are separate questions addressed in Section G.

F. Applications of the Framework

We now apply the framework to settings in which governing references or production conditions change across time. Temporal targets and project state test the selection of evaluative authority. Propagation tests whether a prior error entered the production history of a later response. Retrieval-augmented generation makes the difference between production inputs and governing references especially visible.

F.1 Time, Project State, and Propagation

Although evaluation occurs at a particular interaction time, the time governing an assertion need not be that interaction time. Instead, the task specifies or implies the temporal target to which the assertion should refer. A task may ask for a current officeholder, a historical stock price, or the rule in force on an earlier date. In some cases, the temporal target coincides with the time at which the task is issued; in other cases, it does not. A response accurate at one time may therefore be classified as a hallucination if it is

contradicted by, or lacks required support from, the governing reference applicable to the task’s temporal target.

This temporal distinction also limits what can be inferred from an error. Suppose a model was trained using information available through February. On May 31, a user asks the model in a deployed system for the current price of a stock. The system supplies a price even though it has no access to market information produced after February. If that price differs from the verified May 31 price, the claim may be classified as a hallucination relative to the governing reference. But the classification does not mean that the trained model failed to learn the May 31 price, because that price did not exist in the information used for training. Nor does the classification alone establish why the deployed system supplied the incorrect price.

Project state is another case where the governing reference may change over time. Consider a continuing project in which a team initially discusses one proposal but later rejects it and approves a different course of action. Records of both proposals may remain accessible to the system. If the system is subsequently asked to continue the project, the approved decision—not the superseded proposal—forms part of the authorized project state. A response that continues the project according to the superseded proposal would therefore contain a task-obligation fidelity error. If the response also states that the superseded proposal remains approved, that assertion about the project state would be a hallucination as well. Evaluation of responses in continuing projects should therefore record not only the interaction time and the time relevant to the task, but also the authorized project state as it is updated over time.

Propagation raises a different question from authority selection. Suppose a system misstates a Friday deadline as Monday, retains the incorrect date, and later retrieves it to prepare a schedule. The governing reference still determines that Monday is wrong; propagation concerns the production history through which the earlier error reached the later response. Evidence may include the initial error, the record in which it was retained, the retrieval event, and the later use of the incorrect date.

F.2 Retrieval and Context Construction

Retrieval-augmented generation, or RAG, is a deployment process through which a system retrieves external information and places selected material in the operative context ([12]). It can provide current, private, or task-specific information, but it also makes visible the difference among information available to the system, information presented to the model, and the reference that governs evaluation.

A typical RAG process constructs a query, searches an available corpus, ranks retrieved material, and places selected content in the operative context. Relevant information can be omitted or distorted at any stage: a query may misstate the need, search or ranking may omit an applicable source, and context construction may strip provenance, temporal qualifications, or authority labels. Prior research documents these pipeline failure points ([13], [14]).

Adequate performance at one stage does not establish adequate performance at the next. A relevant document may be retrieved while the needed passage is omitted; an accurate passage may enter the operative context but not be used during generation; and inapplicable material may be retrieved and given

undue weight. These differences help localize an investigation, but they do not determine evaluative authority.

Retrieved material does not become the governing reference merely because it entered the operative context. It may be the governing reference, conflict with that reference, or be irrelevant to the assertion being evaluated. The evaluator must still apply q_i and Π_i to select the governing standard and classify the delivered response. RAG changes how information enters production and where an error may arise, while authority-first evaluation determines how the response should be judged.

G. Discussion and Conclusion

G.1 Consequence and controls

Occurrence of an error does not determine its consequence. The consequence of a hallucination or task-obligation fidelity error depends on the task, the reliance placed on the response, and whether the error is retained or propagated. Consider the same false statement in two settings. In a casual exchange in which no one relies on the statement, the error may have little effect. If the statement is instead entered into a research record, compliance analysis, or organizational decision process, later work may rely on it and extend its effects.

For governance, we therefore use two axes. The actual or potential consequence of an error informs the priority and urgency of the response. Supported production findings inform which controls are plausible matches for the problem. For illustration, evidence that required information never entered the operative context may support changes to retrieval or access, while evidence that accurate information entered the context but was not used may support verification or changes to generation. These examples organize inquiry rather than estimate comparative control effectiveness. Urgent containment need not wait for causal resolution, but a durable response should reflect what the available evidence supports.

Controls can also interact. Retrieval changes may affect which information is salient during generation, and decoding or verification changes may affect whether a system abstains when information is missing. We do not mean that one control is generally ineffective. Rather, a control aimed at one production condition should not be assumed to remedy another without evidence. This interaction is why evaluation and error management should concern a configured system performing a specified task rather than a model name or isolated control considered alone.

G.2 Evaluation sequence

Our analysis can now be brought together as an eight-step sequence for evaluating a configured system that performs a specified task:

1. Specify the task and the applicable authority policy.
2. Identify the assessable assertions and applicable task obligations.
3. Select the governing reference for each assertion and the governing requirement for each task obligation.

4. Classify each assertion-reference relation as supported, contradicted, unsupported, or unresolved, and each obligation as satisfied, violated, or unresolved.
5. Determine whether the response contains hallucination, task-obligation fidelity error, or both.
6. Identify plausible production loci and make causal claims only to the extent supported by production evidence.
7. Assess the consequences of each error, including possible reliance and propagation.
8. Select controls appropriate to the supported findings and the deployment context.

These steps state an analytical order for a complete evaluation, not a mandatory operational chronology. Task and authority specification logically precede classification, and classification precedes claims about cause. In practice, however, a high-consequence error may require immediate containment before causal attribution is complete. The sequence preserves the distinctions needed for analysis without requiring governance action to wait for every later step.

G.3 Limitations

Four principal limitations bound our framework. First, it requires an authority policy but does not supply one. Governing references and resulting classifications are therefore conditional on the policy adopted for the task. The framework organizes that choice, while leaving the evaluator or organization to determine which policy to adopt.

Second, our framework is conceptual. Empirical study is needed to estimate error frequency, establish the relative contribution of production mechanisms, and determine whether the proposed sequence improves evaluator agreement or control selection.

Third, the framework separates consequence assessment from occurrence but does not provide a complete method for measuring adverse effects or selecting controls independently of organizational context. Fourth, localization and causal attribution depend on the availability and completeness of production evidence. Closed or weakly instrumented systems may permit reliable classification of a delivered error while leaving its production history unresolved.

G.4 Conclusion

A response is produced by a configured system but evaluated against governing standards selected for the task. This difference creates the paper’s first problem: production inputs do not necessarily determine evaluative authority. Our authority-first framework addresses it through two parallel branches, comparing assertions with governing references and the delivered response with governing task requirements. Those comparisons support the classifications of hallucination and task-obligation fidelity error.

The second problem is that a valid error classification does not identify how the error arose. We therefore treat localization and causal attribution as later, evidence-dependent inquiries and separate both from consequence and control choice. Applications to time, project state, propagation, and RAG show why these distinctions matter for configured systems. Whether the framework improves evaluation or leads to better-targeted controls remains an empirical question.

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